Front End Assignment

Instructions

Please ensure to go through all instructions before starting the assignment.

1. Thoroughly go through all the requirements and descriptions. Discuss any doubts & queries with the interviewer. You are allowed **10 minutes** to read the instructions.
2. A maximum of **2 hours is allowed for this assignment.**
3. The assignment is marked on the basis of how close your implementation is to the actual requirement in terms of UI and functionality.
4. You are required to implement the assignment using **HTML, CSS and JS (vanilla) only. Bootstrap, Jquery and other libs are not allowed.**
5. Close all the unnecessary applications & tabs.
6. You can use Visual Studio Code or a similar code editor for this assignment.
7. **Googling is permitted for verifying documentation only**. You may refer to MDN, W3Schools for documentation purposes.
8. Use of AI tools like **ChatGPT, deepSeek and github copilot along with any AI extensions is prohibited. Using these will lead to disqualification.**
9. Do not leave the interview without the interviewer's permission.
10. Upon completing the assignment, create a properly compressed zip file following the naming convention [Your Name]-SimonSays-R1.zip and upload it via this link

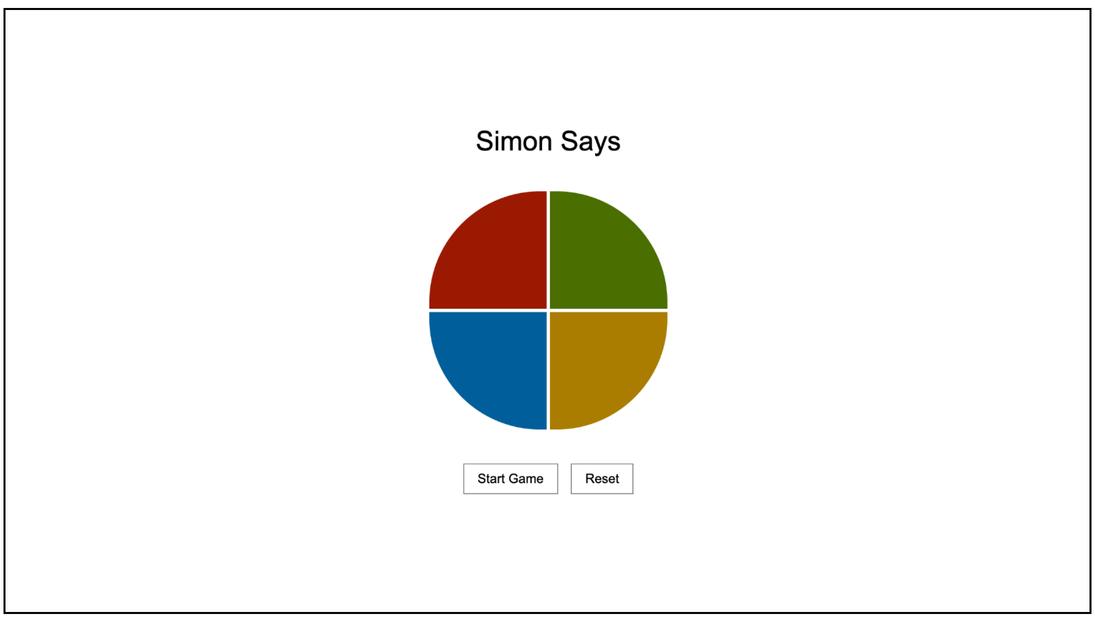
Technical Requirements: Simon Says Game

Look and Setup Requirements

Your app needs to look like the screenshot below.

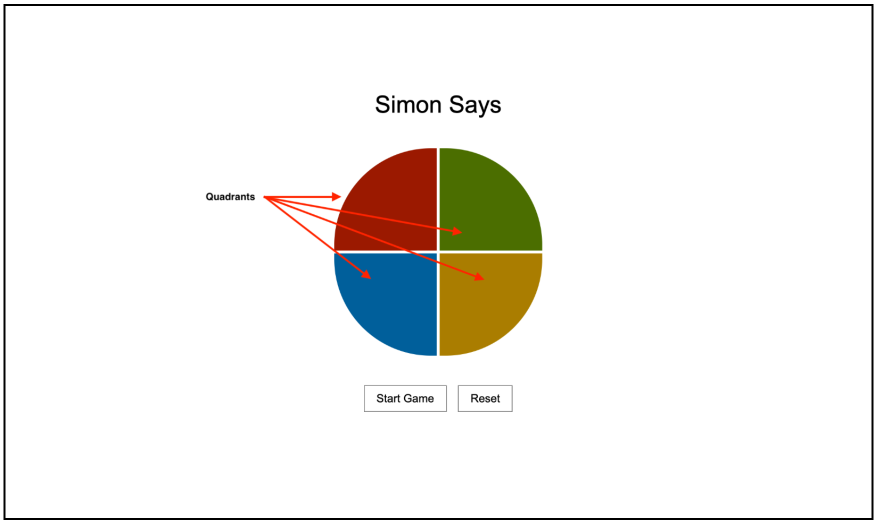
- **Box/Container:** The whole app must be placed in one box in the center of the screen.
- **Structure:** It must have a circle made up of 4 quadrants of Red, Green, Blue, and Yellow color respectively.
- **Buttons:** It should have 2 buttons
  - "Start Game" : Will be used to start the game
  - "Reset" : Will be used to reset the game.

How It Must Work (Step-by-Step)



<u>Game Board</u>

- The application should display a circular game board consisting of four colored quadrants:
  - Red
  - Green
  - Blue
  - Yellow
- Each quadrant should be clickable and should have a blink effect when it is clicked. You may refer to the demo to see how the blinking effect looks like.

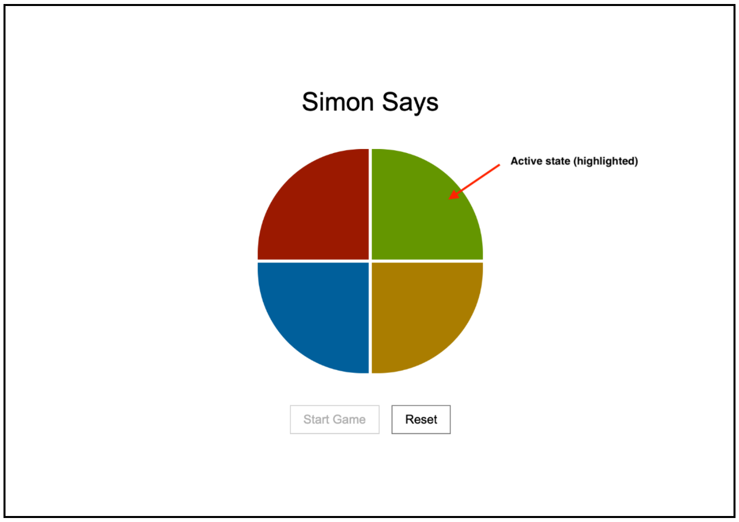


Start Button

- When clicked:
  - A new game begins.
  - The game generates a random color sequence.
- For each color in the sequence:
  - The corresponding quadrant should light-up/blink briefly.
  - There should be a **1 second delay** between consecutive flashes.

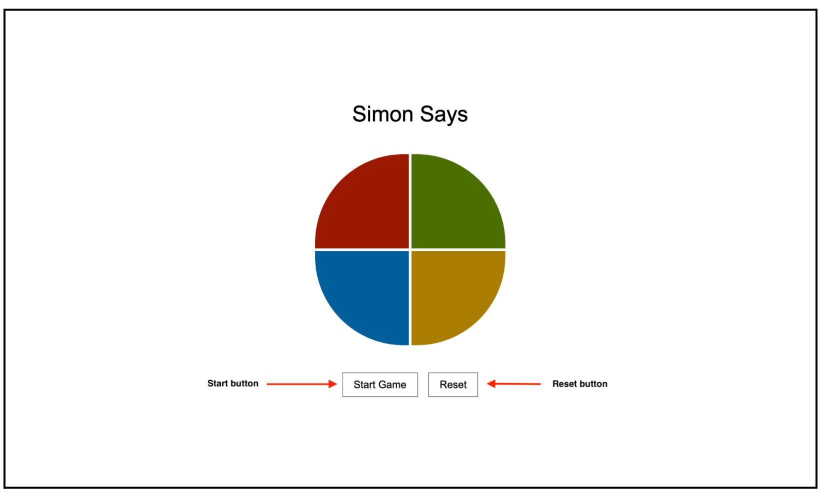
Player's turn

- Once the sequence playback is complete:
  - The user should be able to click the colored quadrants.
  - Clicking a color should provide the flash/blink effect.
  - The sequence is shown to the user by flashing the corresponding colors.
- **Sequence Validation**
  - Each user click should be validated immediately.
  - If the selected color matches the expected color in the sequence, continue accepting input.
  - If the selected color does not match the expected color, show a **"You Lost!"** alert and reset the game.
- **Round Progression**
  - When the user successfully reproduces the entire sequence, the round is completed.
  - A new round begins automatically after a 1 second delay.
  - A new random sequence should be generated for the next round.
  - The sequence starts from 1 flash and goes up to a maximum of 6 flashes. Then it always shows 6 flashes until the player loses.



Reset Game

- Clicking on the **Reset** button should stop the current game immediately.
  - All game data should be cleared.
  - The board should return to its initial state.



You can find a demo video illustrating the entire task along with this document. **All the best :)**